

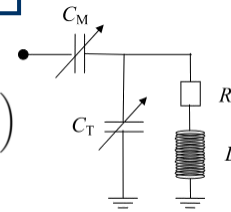
### Complex impedance matching at a specific frequency

$C_T$  sets the resonance frequency:  $\omega_{res} \approx 1/\sqrt{LC_T}$

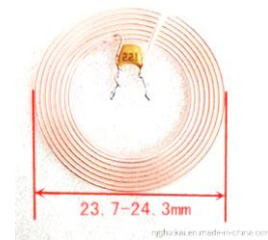
This scales the parasitic resistance up:  $\text{Re } Z_L(\omega = \omega_{res}) \approx R \left( 1 + \frac{L/C_T}{R^2} \right)$

By proper choice of  $L$  and  $C_T$ , the real part of the impedance can be set to 50 Ohm

The imaginary part is inductive, can be compensated by  $C_M$



### Anti-theft LC circuit, inductive coupling

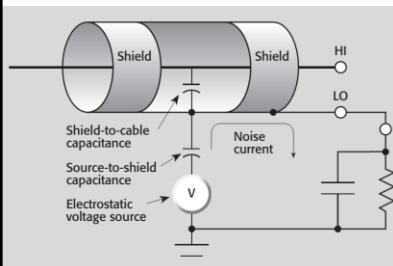
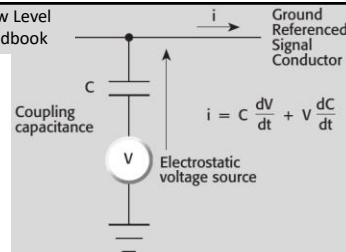


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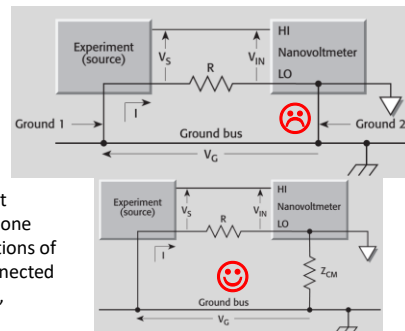
### Suppressing external disturbances: electrostatic coupling, grounding

Our test leads can be capacitively coupled to external voltage sources. If the voltage or the capacitance varies with time (wires moving relative to each other), parasitic currents are generated in the circuit under test, which disturb the measurement. Causes: nearby power lines, RF fields, or even researchers walking past the measurement system :-). This causes unstable measurements, especially in high impedance circuits, where the accumulated charges are slowly discharged.

Source: Keithley Low Level Measurements Handbook



Solution: shield the circuit with a grounded metal enclosure or use shielded conductors. In this case, the currents generated by the electrostatic coupling flow through the shielding to the ground. It is also important that the cables are mechanically fixed, i.e. that the capacitance does not change over time due to mechanical vibrations.



Poor earthing can also cause problems, as there can be a voltage drop between different earth points, resulting in unwanted current flowing through the earth line. Solution: the circuit is grounded at one point (e.g. negative point of voltage source) and the other connections of the earth line (e.g. negative point of voltmeter) are floating or connected to ground through a finite resistor. The use of solid (low resistance, strongly connected) and short earth conductors is recommended.

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### Experiment – Shielding

A 1m BNC coax cable is connected to the input of the scope. We see zero voltage with very little noise. However, if this cable is supplemented with a 1m unshielded cable, we pick up considerable noise. (The shielding of the coax cable in the previous case solved the problem of filtering out this noise.) If the unshielded cable is inserted into a metal box, but the metal box is not grounded, we still see considerable noise, since the potential of the metal box can fluctuate and is transmitted to the cable. If we ground the box directly to the shielding of the coax cable, the noise is eliminated. However, if you ground the box to a completely different point on a long cable, the noise is only reduced, but not eliminated, as there can be a significant voltage drop on the long ground cable.

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### Suppressing external disturbances: Inductive coupling, twisted pairs

Source: Keithley  
Low Level  
Measurements  
Handbook

An unwanted voltage is induced due to a change in the magnetic flux in the area enclosed by the circuit. This can be caused by a changing magnetic field or even by the mechanical movement of the test wires in a static field. This is particularly a problem in low impedance circuits. Solution: mechanical clamping of wires and minimizing the flux around them (e.g. twisted pair wires where opposite fluxes occur at adjacent twists, so that the induced voltages on adjacent sections cancel each other out.)

Example: UTP wire = unshielded twisted pair.

Note: the slowly changing magnetic field is difficult to shield, so a material with a high magnetic permeability (e.g. mu-metal) is used. (At low temperature, the magnetic field is efficiently shielded by a superconductor enclosure.)

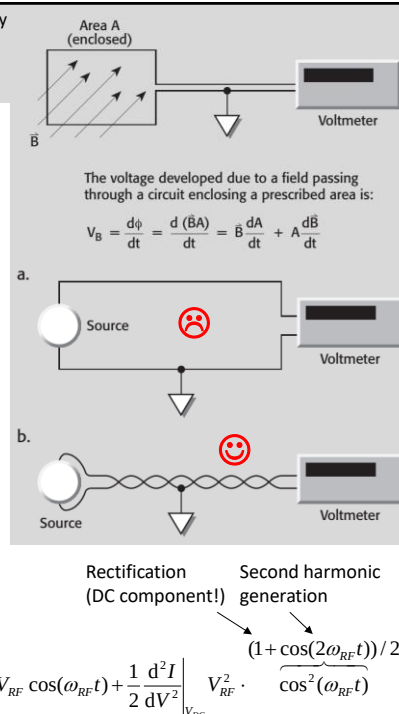
### High frequency disturbance, rectification

Due to the inductive and capacitive coupling, high frequency signals can also cause interference if they are not properly shielded (e.g. radio and TV transmissions, mobile phone WIFI, etc.). These appear as noise at the broadcast frequency (and its harmonics) but can also result in DC voltage in non-linear circuits.

Above 100kHz electrostatic shielding is sufficient, no magnetic shielding is needed. But in case of nonlinear current-voltage characteristics, a radio frequency signal  $V_{RF}\sin(\omega_{RF}t)$

"superimposed" on the  $V_{DC}$  measurement voltage induces a DC component due to the rectification:

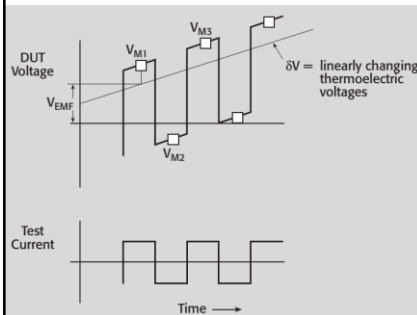
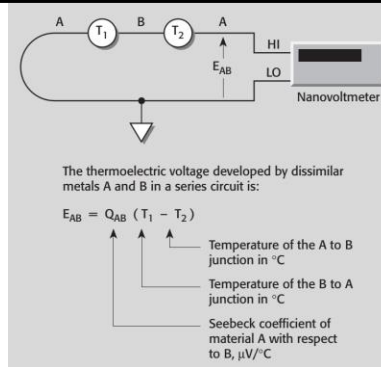
$$I(V_{DC} + V_{RF}\cos(\omega_{RF}t)) \approx I(V_{DC}) + \frac{dI}{dV}\bigg|_{V_{DC}} V_{RF}\cos(\omega_{RF}t) + \frac{1}{2} \frac{d^2I}{dV^2}\bigg|_{V_{DC}} V_{RF}^2 \cos^2(\omega_{RF}t)$$



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### Suppressing external disturbances: thermoelectric voltages, offset compensation

When different metals come into contact, contact potentials occur, which cancel each other out in a closed circuit at a constant temperature. However, if the two contact points of metals A and B shown in the diagram are at different temperatures, a thermoelectric voltage will be induced in the circuit (this is exploited in thermocouple thermometers). At low signal levels, and with significant temperature gradients, the thermoelectric voltage can be compared to the measured signal.



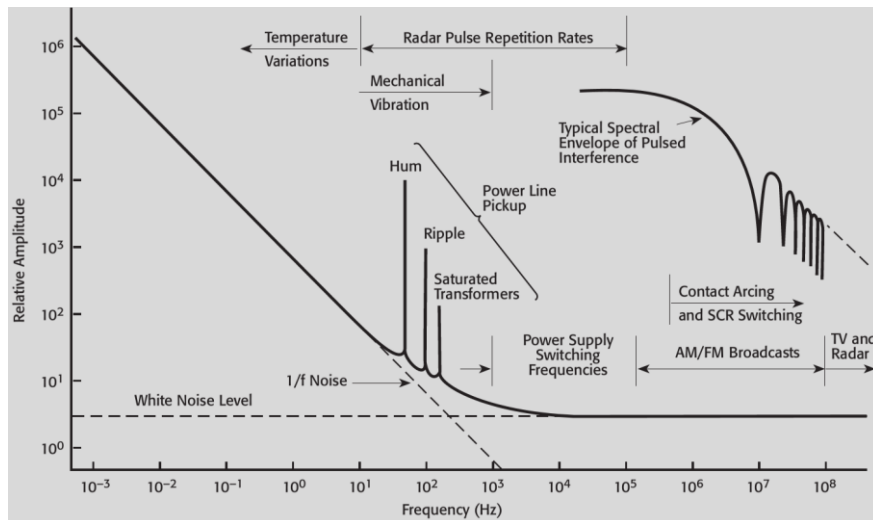
Solution:

- ensure that the contact point pairs of the different metals are at the same temperature
- by varying the polarity of the driving current, the thermoelectric voltage can be compensated, since it does not depend on the current, while the voltage across the resistance under test varies in proportion to the current. The latter is the so-called offset-compensation method.

Source: Keithley Low Level Measurements Handbook

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### Typical frequency spectrum of external disturbance signals



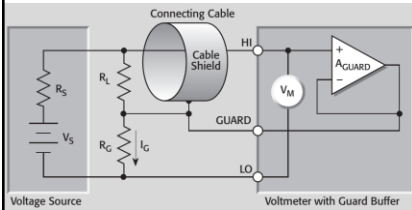
Source: Keithley Low Level Measurements Handbook

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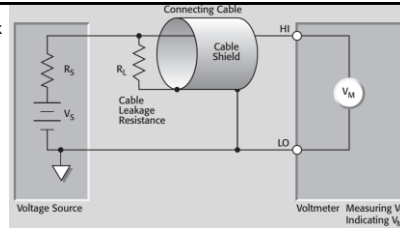
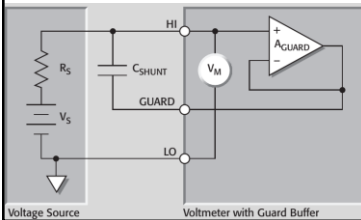
## Guarding

Source: Keithley Low Level Measurements Handbook

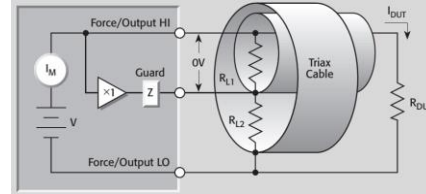
When measuring particularly high impedance circuits, the leakage resistance between the two wires (e.g. inner conductor and shield) can be comparable to the resistance of the circuit, causing significant measurement error.



An even better solution is to use a triaxial cable, i.e. two concentric shields, the inner shielding follows the potential of the middle wire (guarding), while the outer shielding is grounded.



Solution: use a shielded wire, but do not ground the shielding, rather keep it at the potential of the inner wire using a 1x follower amplifier. This way there is no potential difference between the inner core and the shielding, i.e. the leakage current is suppressed. This is called *guarding*.



In a high output impedance circuit, the stray capacitance of the cable is also a problem, as the huge RC time constant means that it takes a very long time to reach a steady state  $V_M$  value after connecting or changing the circuit. This can also be eliminated by guarding, since if the shielding is held at the potential of the inner core, the stray capacitance is not charged, i.e. the circuit responds much faster.